

Simple Transformations

30-Minute Hands-On Review | Week 9 - Section 3C

Learning Objectives: Rename and remove queries and columns using the Power Query Editor | Explain why selecting the correct data type matters for transformations | Change a column data type using the Home ribbon | Describe when and why to add a calculated field | Apply the correct simple transformation to a real analyst scenario

Instructions: Complete all four parts in 30 minutes. Refer to your Section 3C notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term or phrase from Section 3C.

1.	To rename or remove a query or column in Power Query Editor, right-click the item to open the _____.
2.	To change the name of a query (table), right-click the query in the Queries List Pane and select _____.
3.	To delete an unwanted column from a query, select and right-click the column header, then choose _____.
4.	Selecting an appropriate _____ for each column is important because it determines which transformations and aggregations are available.
5.	To change a column's data type using the ribbon, select the column, go to the Home ribbon, and use the Data Type dropdown in the _____.
6.	A new column that is computed from existing column values using a formula expression is called a _____.

Part 2: Matching (~6 minutes)

Write the letter of the correct description next to each term or action.

Term / Action	Description / Example
A. Remove Column	1. The right-click menu that appears when you select a query or column in Power Query Editor
B. Rename Column	2. Replaces an unclear or system-generated column header with a human-readable name
C. Change Data Type	3. Required before applying numeric aggregations like Sum or text operations like Split Column

D. Rename Query	4. Changes the display name of a table as it will appear in the Power BI data model
E. Calculated Field	5. Permanently strips an unwanted column from the query so it is not loaded into the model
F. Context Menu	6. A new column added to a query whose value is derived from a formula based on other columns

Part 3: Put the Steps in Order (~8 minutes)

The steps for four common Power Query transformations are shuffled below. Write the correct order number in the Order column for each transformation group.

Transformation	Order (1-4 or 1-5)	Shuffled Step
Rename a Query		<i>Write the step number in the Order column.</i>
	_____	Select Rename from the context menu
	_____	Open Power Query Editor
	_____	Right-click the query to open the context menu
	_____	Select the query you want to rename in the Queries List Pane
Remove a Column		<i>Write the step number in the Order column.</i>
	_____	Select Remove from the context menu
	_____	Open Power Query Editor
	_____	Select the query that contains the column
	_____	Select and right-click the column header to open the context menu
Rename a Column		<i>Write the step number in the Order column.</i>
	_____	Type the new column name and confirm
	_____	Open Power Query Editor
	_____	Select the query that contains the column
	_____	Select and right-click the column header, then choose Rename
Change a Column Data Type		<i>Write the step number in the Order column.</i>
	_____	Select the desired data type from the dropdown

	_____	Select the column whose type you want to change
	_____	Click the Data Type dropdown in the Transformation Region
	_____	Select the query
	_____	Select the Home ribbon

Part 4: Scenario Analysis (~9 minutes)

Read each scenario. Identify the correct Section 3C transformation to apply and briefly describe the steps you would take in Power Query Editor.

Scenario	Transformation + Steps to Apply
1. A query was automatically named 'Query1' when data was imported. The model will have ten queries and the team needs each to have a meaningful name before publishing the report.	
2. A CSV file has an 'OrderDate' column, but Power Query has detected it as Text type. Time-intelligence calculations in DAX will not work correctly until this is fixed.	
3. An imported table contains 15 columns but the analyst only needs 6 for the report. The remaining 9 columns contain sensitive employee data and should not be loaded into the model.	
4. A column in the source database is named 'cust_nm' due to legacy naming conventions. Report users will see this name on visuals and will find it confusing.	
5. A sales table has a 'Unit Price' column and a 'Quantity' column, but no total revenue column. The analyst wants to add a 'Revenue' column that multiplies the two together before the data reaches the model.	

Bonus / Reflection (if time allows)

Section 3C notes that data sources like text files 'are just guessing data types' and should be reviewed column by column. In 3-5 sentences, explain why this is a risk for data analysis. Give a specific example of a data type mistake and describe the incorrect result it could produce in a Power BI report if left uncorrected.
